

§ Groups in action

Recall $G \curvearrowright X$ means a group G (left) acting on set X . $X \curvearrowright G = G^{\text{op}} \curvearrowright X$ mean right action.

$X \setminus G, X/G$ means $\{G\text{-orbits on } X\}$.

Examples

- $\text{SO}(3) := \{\text{rotations in } \mathbb{R}^3\}$. Consider $\text{SO}(3) \curvearrowright \mathbb{S}^2 := \{v \in \mathbb{R}^3 \mid \|v\| = 1\} \subset \mathbb{R}^3$.
 $v \in \mathbb{R}^3 \setminus \{0\}$, its $\text{SO}(3)$ -orbit $= \{w \in \mathbb{R}^3 \mid \|w\| = \|v\|\}$,
its stabilizer $\text{Stab}(v) := \{g \in \text{SO}(3) \mid gv = v\} = \{R_v(\theta) \mid \theta \in \mathbb{R}/_{2\pi\mathbb{Z}}\}$.
- $H \leq G$, then H acts on G by multiplication in G , with orbits $= Hx$ and stabilizer $= \{1\}$.

Definition $G \curvearrowright X$ is said to be

- **faithful** / 忠实 if $\bigcap_{x \in X} \text{Stab}_G(x) = \{1\}$, i.e. $\forall g \in G$, if $gx = x$ for all $x \in X$, then $g = 1$.
- **transitive** / 传递 if $\forall x, y \in X, \exists g \in G, gx = y$, i.e. X is a single G -orbit.
- **free** / 自由 if $\forall x \in X, \text{Stab}_G(x) = \{1\}$.

Notation $X^G := \{x \in X \mid \forall g \in G, gx = x\} = \{x \in X \mid \{x\} \text{ is a } G\text{-orbit}\}$.

Structure of orbits

Given $G \curvearrowright X \ni x$ and G -orbit Gx , we consider the map

$$\begin{aligned} G &\twoheadrightarrow Gx \\ g &\mapsto gx, \end{aligned}$$

Let $H := \text{Stab}_G(x)$, then $gx = g'x \iff g^{-1}g' \in H \iff g'H \subset gH \iff gH = g'H$.

Hence $G/H \xrightarrow{1:1} Gx$. Moreover $G \curvearrowright G/H \longleftrightarrow G \curvearrowright Gx$ since $a(g, g'H) = gg'H$.

Also $|Gx| = (G : H)$.

Orbit decomposition: $X = \bigsqcup Gx \implies |X| = \sum |Gx| = \sum (G : \text{Stab}_G(x))$ as cardinal numbers.

A useful conclusion in counting is **Burnside lemma**, which is derived from orbit decomposition.

Proposition (Burnside) G group, X non-empty set. Define $X^g = \{x \in X \mid gx = x\}$, then

$$|G \setminus X| \cdot |G| = \sum_{g \in G} |X^g|.$$

Proof Consider a special case where $G \curvearrowright X$ is transitive. From $\text{Stab}_G(gx) = g\text{Stab}_G(x)g^{-1}$ we know $|\text{Stab}_G(gx)| = |\text{Stab}_G(x)|, \forall g \in G$. Hence

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |\text{Stab}_G(x)| = |\text{Stab}_G(x_0)| \cdot \underbrace{(G : \text{Stab}_G(x_0))}_{|X| = \text{length of orbit}} = |G|,$$

which prove the conclusion since $|G \setminus X| = 1$.

For the general case, write $X = \bigsqcup Gx_i, 1 \leq i \leq n$. Then $G \curvearrowright X_i$ is transitive for all i , hence

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |\text{Stab}_G(x)| = \sum_{i=1}^n \sum_{x \in X_i} |\text{Stab}_G(x)| = n|G| = |G \setminus X||G|.$$

□

In other words, each element in G has “around $|G \setminus X|$ ” fixed points in X .

Remark Given G group, X non-empty set, we have

$$\{a : G \times X \rightarrow X, \text{ action } G \curvearrowright X\} \longleftrightarrow \{\text{homomorphisms } A : G \rightarrow S_X\},$$

where $S_X := \{\text{bijections } X \rightarrow X\}$.

Indeed: to give $a : G \times X \rightarrow X$ amounts to $A : G \rightarrow \{\text{maps } X \rightarrow X\}$ s.t. $A(g)(x) = a(g, x)$.

$$\begin{aligned} a(g_1 g_2, x) &= a(g_1, a(g_2, x)), \forall x \in X \\ \iff A(g_1 g_2) &= A(g_1) A(g_2). \end{aligned}$$

$$a(1_G, x) = x, \forall x \in X \implies A(\text{id}_G) = \text{id}_X.$$

So:

$$\begin{aligned} \{G \curvearrowright X\} &\longleftrightarrow \{A : G \rightarrow \{\text{maps } X \rightarrow X\}\} \text{ s.t. } \begin{cases} A(1_G) = \text{id}_X \\ A(g_1 g_2) = A(g_1) A(g_2) \end{cases} \\ &= \{\text{homomorphisms } A \rightarrow S_X\}. \end{aligned}$$

□

$G \curvearrowright X$ is faithful $\iff (\forall g \in G, A(g) = \text{id}_X \iff g = 1_G) \iff A$ is injective,
 $\therefore A(g_1) = A(g_2) \implies A(g_1^{-1} g_2) = \text{id}_X$.

Theorem (Cayley) Every group G can be embedded into S_G .

Proof Let G acts on itself by left multiplication, i.e. $a(g, x) = gx$.
 $\implies$ every finite group G embeds into some S_n , ideally $n \ll |G|$.

□

§ Case study: permutations and cycles

Let X non-empty set, $m \geq 1, a_1, \dots, a_m \in X$ distinct, say indexed by $\mathbb{Z}/m\mathbb{Z} \longleftrightarrow 1, \dots, m$.

Permutations $\sigma \in S_X$ of the form

$$\begin{cases} \sigma(a_i) = a_{i+1}, & i \in \mathbb{Z}/m\mathbb{Z} \\ \sigma(x) = x, & \forall x \in X \setminus \{a_1, \dots, a_m\} \end{cases}$$

are called m -cycles in S_X .

Notation $\sigma = (a_1 \dots a_m) \in S_X$.

Special cases: $m = 1 \implies \text{id}_X, m = 2 \implies (a_1 \ a_2)$ tranposition.

We say two cycles $(a_1 \dots a_m), (b_1 \dots b_n)$ are **disjoint** if $\{a_1, \dots, a_m\} \cap \{b_1, \dots, b_n\} = \emptyset$.

Easy observations

- $\text{ord}(a_1 \dots a_m) = m$ in S_X ,
- σ, τ disjoint $\implies \sigma\tau = \tau\sigma$.

Hereafter $X = \{1, \dots, n\}, S_X = S_n$.

Proposition Every $\sigma \in S_n$ decomposes into pairwise disjoint cycles

$$\sigma = (a_{1,1} \dots a_{1,l_1}) \dots (a_{m,1} \dots a_{m,l_m})$$

where $\sum l_i = n$, and such m and cycles are uniquely determined by σ up to permutations.

Proof $\langle \sigma \rangle \curvearrowright \{1, \dots, n\} \implies$ decomposition into $\langle \sigma \rangle$ -orbits $C_1 \sqcup \dots \sqcup C_m$.

$\forall 1 \leq i \leq m$, pick $a_{i,1} \in C_i$ and set $a_{i,k} = \sigma^{k-1}a_{i,1}$. Take the minimal $l_i \geq 1$ s.t. $\sigma^{l_i}a_{i,1} = a_{i,1}$, $\implies \sigma \curvearrowright C_i$ by $(a_{i,1} \dots a_{i,l_i})$, $\because 1 \leq h < k \leq l_i, a_{i,h} \neq a_{i,k}$, otherwise $\sigma^{k-h}a_{i,1} = a_{i,1}$.

Different choice of $a_{i,1}$ is equivalent to cyclically permuting $a_{i,1}, \dots, a_{i,l_i}$.

Hence σ acts on $C_1 \sqcup \dots \sqcup C_m$ as $(a_{1,1} \dots a_{1,l_1}) \dots (a_{m,1} \dots a_{m,l_m})$, with $\sum l_i = \sum |C_i| = n$. $\square$

Convention Omit the 1-cycles in the cycle decomposition.

Easy observations

- $\text{ord}(\sigma) = \text{lcm}(l_1, \dots, l_m)$.

$$\because \forall k \in \mathbb{Z}, \sigma^k = (a_{1,1} \dots a_{1,l_1})^k \dots (a_{m,1} \dots a_{m,l_m})^k,$$

$$\begin{aligned} \sigma^k = \text{id}_n &\iff (a_{i,1} \dots a_{i,l_i})^k = \text{id}, \forall 1 \leq i \leq m \\ &\iff l_i \mid k, \forall i. \end{aligned}$$

- $\text{sgn}(\sigma) = (-1)^{\sum (l_i-1)}$, $\because \text{sgn} : S_n \rightarrow \{\pm 1\}$ is a homomorphism $\implies$ suffices to show

$$\text{sgn}((a_1 \dots a_l)) = (-1)^{l-1}.$$

- $l = 2$ case: $\text{sgn}((a_1 a_2)) = -1$.

- $l > 2$ case: $(a_1 \dots a_l) = (a_1 a_l)(a_1 \dots a_{l-1})$. Check inductively. $\square$

§ Applications of orbit decomposition

Idea Counting.

Definition G finite group, p prime number. If $|G| = p^n, n \in \mathbb{Z}_{\geq 0}$, we say G is a p -group.

Proposition G p -group, $G \curvearrowright X$ finite set, then $|X^G| \equiv |X| \pmod p$.

Proof $X = Gx_1 \sqcup \dots \sqcup Gx_m, H_i := \text{Stab}_G(x_i) \implies |X| = \sum (G : H_i)$. Here

$$(G : H_i) = \frac{|G|}{|H_i|} = \begin{cases} H_i \neq G, & \in p\mathbb{Z} \\ H_i = G, & 1 \end{cases}$$

Hence $|X| \equiv |\{i \mid H_i = G\}| \pmod p = \text{"the number of singleton } G\text{-orbits } \subset X" = |X^G|$. $\square$

Proposition G p -group, $G \neq \{1\}$. Then its center $Z_G \neq \{1\}$.

Proof Define an action $a : G \times G \rightarrow G, (g, x) \mapsto gxg^{-1}$.

$x \in G$ is a fixed point $\iff gxg^{-1} = x, \forall g \in G \iff gx = xg, \forall g \in G \iff x \in Z_G$.

So $|Z_G| = |\{G\text{-fixed points}\}| \equiv |G| \pmod p \equiv 0 \pmod p, 1 \in Z_G \implies |Z_G| > 1$. $\square$

§ Normal subgroups and quotient groups

Let G group, $S \subset G$. For $x, y \in G$, define $xSy = \{xsy \mid s \in S\} \subset G$.

Definition A subgroup $H \leq G$ is said to be **normal** if $\forall g \in G, gHg^{-1} = H$, denoted as $H \triangleleft G$.

Equivalently:

- $gHg^{-1} \subset H, \forall g \in G, \because \text{Put } g^{-1} \implies g^{-1}Hg \subset H \implies H \subset gHg^{-1}$.
- $gH = Hg, \forall g \in G$.
- $gH \subset Hg, \forall g \in G$.

Examples

- G abelian $\iff \forall H \leq G, H \triangleleft G$.
- $\{1\}, G \triangleleft G$.
- $Z_G \triangleleft G, \because gZ_G = Z_Gg$.
- $N \triangleleft G, H \leq G$, then $N \cap H \triangleleft H$.
- $(G : H) = 2 \implies H \triangleleft G, \because G = H \sqcup (G \setminus H = gH = Hg)$ in terms of left and right cosets.